

Question 1: What is a variable in C/C++?

Answer: A variable is a named memory location used to store data.

Question 2: Why do we use variables in programs?

Answer: Variables are used to store values that can change during program execution.

Question 3: What is the difference between a variable name and its value?

Answer: A variable name identifies the memory location, while its value is the data stored in that location.

Question 4: Which of the following is a valid variable name? a) 2num b) total_marks c) for d) my marks

Answer: b) total_marks

Question 5: Why is Age different from age?

Answer: C++ is case-sensitive, so Age and age are treated as different variables.

Question 6: Can a variable name contain spaces? Why?

Answer: No, variable names cannot contain spaces because spaces separate words in programming.

Question 7: Can a variable name start with an underscore (_) ?

Answer: Yes, a variable name can start with an underscore.

Question 8: What happens if two variables have the same name in the same scope?

Answer: A compilation error occurs because duplicate variable names are not allowed in the same scope.

Question 9: Write three meaningful variable names for storing student information.

Answer: studentName, studentAge, studentMarks

Question 10: Which naming convention is better and why: studentMarks or sm?

Answer: studentMarks is better because it is meaningful and easy to understand.

Question 11: What is a data type?

Answer: A data type specifies the type of values a variable can store.

Question 12: Which data type is used to store whole numbers?

Answer: int

Question 13: Which data type is used to store decimal values?

Answer: float or double

Question 14: What is the difference between float and double?

Answer: double provides more precision and can store larger decimal values than float.

Question 15: Which data type stores a single character?

Answer: char

Question 16: Which data type stores True/False values?

Answer: bool

Question 17: What data type is suitable for storing a person's name?

Answer: string

Question 18: Predict the output: int age = 18; cout << age;

Answer: 18

Question 19: Identify the data type: 25, 3.14, 'A', true.

Answer:

- 25 → int
- 3.14 → float/double
- 'A' → char
- true → bool

Question 20: What happens if you store a decimal number in an int variable?

Answer: The decimal part is discarded.

Question 21: What is the purpose of cin?

Answer: cin is used to take input from the user.

Question 22: What is the purpose of cout?

Answer: cout is used to display output on the screen.

Question 23: Which symbol is used with cin?

Answer: >>

Question 24: Which symbol is used with cout?

Answer: <<

Question 25: Explain: cin >> age;

Answer: It reads a value entered by the user and stores it in the variable age.

Question 26: Explain: cout << age;

Answer: It displays the value of the variable age.

Question 27: What is the purpose of endl?

Answer: endl moves the cursor to the next line.

Question 28: Write a statement to input a student's marks.

Answer:

```
cin >> marks;
```

Question 29: Write a statement to display "Welcome to C++".

Answer:

```
cout << "Welcome to C++";
```

Question 30: Predict the output: `cout << "Programming";`

Answer:

Programming

Question 31: What is an operator?

Answer: An operator is a symbol that performs operations on data.

Question 32: Which operator is used for addition?

Answer: +

Question 33: What is the result of `10 % 3`?

Answer: 1

Question 34: What is the result of `8 % 2`?

Answer: 0

Question 35: Which operator checks equality?

Answer: ==

Question 36: What is the difference between = and ==?

Answer: = assigns a value, whereas == compares two values.

Question 37: What is the result of `5 > 3`?

Answer: true

Question 38: What does && represent?

Answer: Logical AND.

Question 39: What does || represent?

Answer: Logical OR.

Question 40: What does ! represent?

Answer: Logical NOT.

Question 41: Write a program to print "Hello World".

Answer:

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello World";
    return 0;
}
```

Question 42: Write a program to input two numbers and display their sum.

Answer:

```
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cin >> a >> b;
    cout << "Sum = " << a + b;
    return 0;
}
```

Question 43: Write a program to calculate the area of a rectangle.

Answer:

```
#include <iostream>
using namespace std;

int main() {
    int length, width;
    cin >> length >> width;
    cout << "Area = " << length * width;
```

```
    return 0;
}
```

Question 44: What formula is used to calculate Simple Interest?

Answer:

$$SI = \frac{P \times R \times T}{100} \quad SI = 100P \times R \times T$$

Question 45: Write a program to calculate Simple Interest.

Answer:

```
#include <iostream>
using namespace std;

int main() {
    float P, R, T, SI;
    cin >> P >> R >> T;
    SI = (P * R * T) / 100;
    cout << "Simple Interest = " << SI;
    return 0;
}
```

Question 46: Why is a temporary variable used while swapping two numbers?

Answer: A temporary variable stores one value safely so that it is not lost during swapping.

Question 47: Swap the values: a = 10, b = 20.

Answer:

After swapping:

- a = 20
- b = 10

Question 48: What formula is used to convert Celsius into Fahrenheit?

Answer:

$$F = \frac{9}{5}C + 32 \quad F = 59C + 32$$

Question 49: What is the Fahrenheit value of 0°C?

Answer: 32°F

Question 50: Why can integer division give incorrect results in the Celsius-to-Fahrenheit program?

Answer: Integer division truncates decimal values. For example, 9/5 becomes 1, giving incorrect results. Use 9.0/5 or 1.8 instead.